FROM MICROBIAL ABUNDANCE TO WAYPOINT-6M EMBEDDINGS AND ASD CLASSIFICATION

A complete, step-by-step explanation of the analyses already performed, how taxonomic tokenization works, what the results mean, and the next fine-tuning experiments.

PROJECT	DATASET	CURRENT STAGE
ASD versus Healthy from gut microbiome	389 runs × 1,068 initial features	Frozen Waypoint-6m + four classifiers completed

Key verified numbers

Input	Waypoint preparation	Embedding	Prediction
389 × 1,068	389 × 355	389 × 256	Random stratified 10-fold CV

Samples: 208 ASD and 181 Healthy. **Vocabulary:** 229 of 355 prepared taxonomic groups recognized. **Best frozen-embedding model by AUC:** SVM (RBF), 0.6715 ± 0.0801. **Prior full-feature XGBoost AUC:** 0.7911 ± 0.0551.

Scope and evidence: This is an explanatory record of the uploaded scripts, original/corrected NPZ files, command outputs and user-reported model results. It labels completed work separately from proposals. The 46 source-duplicate corrections are based on an archived audit; independent examination of the original source records remains outstanding.

Reading guide: Pages 2–9 explain the completed workflow. Pages 10–12 interpret results and limitations. Pages 13–15 explain the future methods and how to run a defensible comparison.

1. Complete workflow at a glance

This is the actual path used for the frozen Waypoint experiment. The arrows represent data transformations, not separately trained models.



Not part of this executed path: AGORA2 flux/metabolite calculations, ASD-specific Waypoint fine-tuning, LoRA and full transformer retraining.

2. Original data and duplicate-abundance review

2.1 What the 389 × 1,068 matrix means

Each **row** is a sequencing run (identified by Run ID). Each **column** contains the relative abundance of a named taxon. These are microbial abundance features, not gene-expression counts and not image features.

Original taxon type	Columns	What they represent
Species named	829	Species-labelled abundance fields; 827 unique species TaxIDs
Genus named	237	Abundance already reported at the genus level
Subspecies	1	Mapped using available ancestry
Unknown	1	Not mapped to a named taxon in the Waypoint input
Total	1,068	All original abundance feature columns

2.2 What was wrong with some source entries?

The archived audit flagged **46 groups** of repeated abundance source records across **45 runs**. These were suspected instances of counting the same measurement twice, *not* duplicated people, duplicated run IDs, or duplicate TSV header columns.

Real example	Existing matrix value	Candidate correction
SRR13327425 – [Clostridium] methylpentosum	0.01610%	0.00805%

A corrected TSV and a corrected NPZ were generated while retaining all 389 runs, 1,068 taxonomic columns and outcome labels. Row totals after the corrections are approximately 100%, between 99.99992% and 100.00008%.

Critical qualification: These corrections were validated against saved audit records and row-sum consistency. The original unaggregated source rows were not independently rechecked; therefore, call the changes *audit-supported candidate corrections* in a manuscript until that provenance check is complete.

3. Why 1,068 microbial features became 355 taxonomy groups

Waypoint-6m is genus-oriented. We do not feed 827 species separately as if every species had its own pretrained token. First we combine original abundance columns according to NCBI taxonomic lineage.

3.1 Species-to-genus aggregation

For each original taxon, the stored NCBI taxonomy lineage is consulted. If the organism has a genus ancestor, its abundance is added to the appropriate genus group. Existing genus abundance fields are grouped consistently with this mapping.

Example (illustrative)	Abundance	Mapped output
Bacteroides fragilis	8%	Bacteroides
Bacteroides thetaiotaomicron	5%	Bacteroides
Bacteroides uniformis	3%	Bacteroides
Final combined Bacteroides group	16%	One genus column

3.2 Exactly how the 355 groups arose

Category	Distinct groups	Description
Resolved genus groups	340	234 distinct genus groups from genus-named columns, plus 106 additional genera found through species ancestry
Higher-rank groups	15	9 families, 2 orders, 1 class and 3 phyla lacking genus resolution
Waypoint matrix	355	340 + 15; kept for all 389 runs

3.3 Relative-abundance fractions

For each run, usable named abundance is divided by that run's total **usable resolved named abundance**. The generated Waypoint input has row sums near 1. For instance, if a genus has 16% and the usable resolved mass is 80%, its Waypoint fraction is 16/80 = **0.20**.

The original **Unknown** column and four unresolved taxonomic source features are retained for auditing but do not form one of the 355 resolved Waypoint groups. No ASD labels are used in genus aggregation.

Important: The 355 taxonomic groups are still abundance features at this stage. They are *not* the 256 transformer embedding dimensions that appear later.

4. How Waypoint vocabulary matching was performed

The checkpoint **outpost-bio/Waypoint-6m** supplies a taxonomic vocabulary (`vocab.json`) and a custom taxonomy tokenizer. Once Hugging Face access was approved, the earlier preparation script downloaded and checked these against all 355 prepared taxonomic paths.

Vocabulary-recognized groups 229 / 355 (64.5%)

Rank	Prepared	Recognized	Unrecognized
Genus	340	219	121
Family	9	8	1
Order	2	1	1
Class	1	1	0
Phylum	3	0	3
TOTAL	355	229	126

4.1 What happens to an available token?

Waypoint extracts the relevant rank, usually the genus part (such as `g__Alistipes`), and uses the checkpoint vocabulary to look up a numerical token ID. This number is a dictionary index, not a concentration or a disease label.

Taxon	Vocabulary ID	Recognized?
g__Alistipes	4248	Yes
g__Bacteroides	4824	Yes
g__Phocaeicola	10026	Yes
g__Segatella	1 /	No

4.2 What happened to the other 126?

They remained in the **389 × 355** input matrix and in the corrected NPZ. During the **official embedding conversion**, taxa whose ID mapped to <unk> were skipped. Thus, they did **not** contribute their own token to the transformer. They were not replaced silently with another genus.

Scientific nuance: 229/355 is the number of potential groups recognized across the cohort. Each particular run contains only some of these groups. Missing taxa in the vocabulary can remove a large fraction of abundance in an individual sample.

5. Worked real example: SRR10338792

This run appears in the corrected NPZ and embedding CSV. We will follow it through abundance normalization and tokenizer lookup. The numbers below are extracted from the corrected archive.

Starting sample	Value
Named microbial abundance	78.86719%
Unknown abundance	21.13281%
Nonzero prepared taxonomy groups	72
Groups recognized by Waypoint	50
Groups absent from its vocabulary	22

5.1 Original percentage to prepared fraction

Genus	Original abundance	Normalized named fraction	Token ID
Alistipes	34.01851%	0.431339	4248
Bacteroides	13.15917%	0.166852	4824
Phocaeicola	11.04380%	0.140030	10026
Simiaoa	5.21420%	0.066114	1 / skipped
Dialister	2.84711%	0.036100	6441

Example calculation: Alistipes normalized abundance = 34.01851 ÷ 78.86719 ≈ **0.431339**. The model's original taxonomic lineage is used to obtain its genus token `g__Alistipes`, then ID 4248.

5.2 What is lost for this particular sample?

Recognized proportion of usable named abundance 90.36 / 100 (90.4%)



The recognized groups account for **90.36%** of usable prepared named abundance and the 22 unrecognized groups for **9.64%**. Because this run also has 21.13% Unknown, the represented abundance is approximately **71.27%** of its original total profile.

Example of a skipped genus: Simiaoa has 6.61% of the prepared named abundance but resolves to unknown token ID 1, so the official embed routine excludes it from the token sequence. This can remove substantial information.

6. From token IDs to the actual Waypoint input sequence

6.1 Vocabulary conversion

The recognized taxa are represented by integers, for example Alistipes → 4248, Bacteroides → 4824 and Phocaeicola → 10026. A numeric ID itself does not encode abundance. The taxon labels remain linked to their abundance fractions during the preparation stage.

6.2 Ordering by abundance statistics

Your run downloaded the checkpoint file `token_std_means.parquet` and printed **14,389 token statistics**. The official embedding implementation can order recognized tokens using abundance **z-scores** rather than simply sorting them by largest observed abundance.

> **Abundance z-score:** z(taxon) = [sample abundance − checkpoint reference mean] ÷ checkpoint reference standard deviation. For Alistipes, sample abundance is 0.431339. The reference mean and SD come from the pretrained checkpoint statistics, not the ASD labels.

For example, a taxon at 3% could have a larger z-score than a taxon at 15% if the first is unusually high relative to its reference distribution. This is why the model can see a different sequence order from a descending-abundance table.

6.3 Constructing the transformer sequence



A previous explanation illustrated BOS/EOS and padding to a maximum length of 512. The exact sorted token list, padding layout and position of special tokens for this run should be verified by inspecting the installed tokenizer output; those exact details are **not** established solely by the NPZ.

7. What pretrained Waypoint-6m does internally

Waypoint-6m is a GPT-2-style causal transformer pretrained on taxonomic sequences. The checkpoint provides existing learned weights; the ASD run did not start those weights from scratch.

Component	Role
Taxonomic tokenizer	Converts known taxonomic labels to integer IDs
Input token embeddings	Maps each integer token into a vector of hidden features
Positional information	Allows the model to distinguish token order
8 transformer blocks	Learn progressively context-dependent microbial representations
4 attention heads	Allow multiple learned attention patterns within each block
Hidden width 256	Each final token state contains 256 numerical dimensions
last_token pooling	Returns one sample-level vector based on the last valid token state

7.1 Why the result is 256, not 355 or 1,068

The number **256** is fixed by Waypoint's hidden width. It is not a chosen count of the "best" 256 species. Each of the 256 dimensions is a learned mathematical feature, potentially reflecting relationships among multiple taxa in the ordered community.

7.2 What the GPU run established

```
Loading model from outpost-bio/Waypoint-6m ...
Loading weights: 100%
Using device: cuda
Loaded token_std_means (14389 tokens)
Embedding: 13/13 batches
Saved embeddings of shape (389, 256)
```

Your output CSV contains **389 rows × 257 columns**: one `Sample ID` column and 256 embedding columns `dim_0` through `dim_255`. Your validation found no missing values and no duplicate sample IDs.

Interpretation: Embedding dimensions are not ASD probabilities, species counts, measured metabolites or direct evidence of microbial causality. They are features extracted from the pretrained model.

8. Frozen Waypoint embeddings: the completed ML experiment

The transformer was used once to extract all 389 embeddings. Its weights were not updated for ASD. Then four conventional classifiers were trained on the 256-dimensional vectors.

8.1 How random stratified 10-fold evaluation works

All 389 runs are shuffled reproducibly into 10 approximately class-balanced folds. In each iteration, nine folds are used for training and one fold for testing. Across the 10 iterations, each run is in the test fold once. The model is fitted anew for each fold.

```
StratifiedKFold(n_splits=10, shuffle=True, random_state=42)
```

Classifier	Settings used	StandardScaler?
XGBoost	Binary logistic objective; baseline defaults; seed 42	No
Logistic Regression	Default solver/regularization; max_iter = 5000	Yes
SVM (RBF)	Default RBF SVC	Yes
Random Forest	Default forest settings; seed 42	No

Scaling for SVM and logistic regression is **fitted inside the training fold** and then applied to the held-out fold. No hyperparameter tuning and no species feature selection were performed. The same random stratified folds were used for all four classifiers.

8.2 Six reported metrics

Accuracy: overall correct fraction. **AUC:** ranking ability across thresholds. **Sensitivity:** proportion of ASD runs detected. **Specificity:** proportion of Healthy runs correctly identified. **Precision:** fraction of positive calls that are ASD. **F1:** harmonic mean of precision and sensitivity.

Each metric is summarized as its **mean ± sample standard deviation** across 10 test folds; SD indicates fold-to-fold variability, not a confidence interval.

9. Actual frozen-embedding classification results

These are the four rows you supplied after running the 10-fold script on the GPU server. Values are fractions, not percentages.

Classifier	Accuracy (mean ± SD)	AUC (mean ± SD)
XGBoost	0.6148 ± 0.0798	0.6551 ± 0.0680
Logistic Regression	0.5812 ± 0.0863	0.5857 ± 0.1073
SVM (RBF)	0.6274 ± 0.0779	0.6715 ± 0.0801
Random Forest	0.6121 ± 0.0676	0.6483 ± 0.0532



All remaining metrics

Classifier	Sensitivity	Specificity	Precision	F1
XGBoost	0.6212 ± 0.0989	0.6082 ± 0.0968	0.6464 ± 0.0778	0.6314 ± 0.0802
Logistic Regression	0.5919 ± 0.1317	0.5687 ± 0.0650	0.6061 ± 0.0773	0.5969 ± 0.1048
SVM (RBF)	0.6550 ± 0.1159	0.5968 ± 0.1009	0.6511 ± 0.0715	0.6498 ± 0.0853
Random Forest	0.6883 ± 0.0732	0.5251 ± 0.1067	0.6275 ± 0.0633	0.6548 ± 0.0566

Result: SVM (RBF) had the highest mean accuracy (0.6274) and AUC (0.6715). Random Forest had slightly higher F1 (0.6548). The best model depends on the metric chosen; none establishes clinical prediction utility.

10. Compare Waypoint with the original abundance baseline

Your prior full-feature XGBoost experiment used the same corrected cohort size and a random stratified 10-fold setup (seed 42). It is a useful baseline comparison, but statistical uncertainty about the performance difference still requires a formal paired analysis.

Representation + classifier	Features	Accuracy	AUC
Corrected original abundance + XGBoost	1,068	0.7405 ± 0.0383	0.7911 ± 0.0551
Frozen Waypoint + XGBoost	256	0.6148 ± 0.0798	0.6551 ± 0.0680
Frozen Waypoint + SVM (RBF)	256	0.6274 ± 0.0779	0.6715 ± 0.0801

On these reported random-fold comparisons, the original abundance representation outperformed the frozen Waypoint embeddings. This does **not** prove that Waypoint cannot work for ASD; it shows that the current frozen embedding representation and default downstream classifiers did not improve these metrics.

Why might frozen Waypoint underperform?

• **Out-of-vocabulary taxa:** 126 of 355 prepared groups are not directly recognized; highly abundant taxa such as Segatella can be skipped.

• **Different input encoding:** the transformer uses taxonomic token sequences ordered by abundance statistics rather than preserving every raw abundance value as an independent numerical feature.

• **Training-domain differences:** the patterns learned during broad microbiome pretraining may not align with ASD-related variation in this cohort.

• **Small, heterogeneous cohort:** the 389 observations span multiple projects and sequencing preparations; random folds do not ensure generalization to a new study.

Do not claim a cause without testing it: The vocabulary mismatch is a plausible limitation, not a demonstrated explanation for the lower AUC. Report sample-level coverage and test its association with held-out prediction errors.

11. How to make the comparisons scientifically defensible

11.1 Data splitting and leakage

Random 10-fold CV is the completed exploratory benchmark. Related sequencing libraries or samples from the same biological subject may be split across training and testing. Consequently, a separate **grouped** or **project-held-out** evaluation is important before asserting cross-study generalization.

11.2 Coverage metrics should accompany AUC

Coverage measure	Value / meaning
Global vocabulary group coverage	229 / 355 = 64.5% of candidate taxonomy groups
Median recognized fraction of named abundance	92.23%
Median recognized fraction of full original profile	82.66%, includes Unknown in denominator
Lowest known-named coverage observed	About 14.34% for an individual run
Example run SRR10338792	50 of its 72 nonzero groups recognized

The 229 group count is not a per-run count. A taxon may exist in the vocabulary but be absent from a given microbiome sample. Missingness is also uneven: samples dominated by an unrecognized genus can lose much more signal.

11.3 Additional manuscript checks

- Recheck the 46 corrected source-record groups against original unaggregated abundance records, if accessible.
- Calculate metrics by held-out project and by microbiome preparation type, not only pooled random folds.
- Retain per-fold test predictions for a paired comparison with the original-abundance baseline.
- Repeat model validation with frozen hyperparameters and truly independent grouping when available.
- Avoid describing microbial taxa or Waypoint embeddings as causal gut–brain mechanisms without complementary functional or clinical data.

12. The five ways of adapting or using Waypoint

The key choice is **which parameters receive gradient updates**. “Pretrained” means the backbone starts from Waypoint-6m checkpoint weights, not random weights.

Method	Trainable parameters	What you pass into training	Status
1. Frozen embeddings + conventional ML	Separate downstream XGBoost, SVM, LR, RF	Saved 389 × 256 embedding vectors	Done
2. Frozen backbone + MLP head	New neural classification head only	Embeddings or model pooled outputs	Planned
3. LoRA adaptation + classifier	Small low-rank adapter weights plus head; base weights frozen	Original tokenized microbiome sequences	Proposed next
4. Partial transformer fine-tuning	Chosen late transformer blocks and head	Original tokenized sequences	Planned
5. Full transformer fine-tuning	All backbone weights and new classification head	Original tokenized sequences	Planned

What “freezing” means in one sentence

A frozen parameter contributes to a forward prediction but is **not updated by backpropagation**. A trainable parameter can change to reduce the ASD-versus-Healthy training loss.

What “head” means

A classification head converts the 256-dimensional sample representation into a single score or logit. Applying a sigmoid yields an estimated ASD-class probability. The new head must be trained; it does not come automatically from Waypoint’s generic language-model pretraining objective.

13. Detailed comparison: MLP, LoRA, partial and full fine-tuning

Method 2: Frozen backbone with a neural head

Keep Waypoint weights fixed. Use its 256-number vector as input to a small trainable neural network. Compared with Logistic Regression, a non-linear MLP can learn interactions among embedding dimensions, but can also overfit a 389-run cohort. The previously completed four classifiers are not this neural-head experiment.

Method 3: Low-rank adaptation (LoRA)

LoRA inserts small trainable low-rank matrices into selected layers while keeping the main pretrained matrices frozen. For a weight matrix W, an adapted operation has the conceptual form **W + ΔW**, where **ΔW = BA** and A/B have much smaller rank than W. Train these adapters and a classification head from ASD/Healthy labels.

Advantages: far fewer trainable parameters than full fine-tuning and usually reduced memory needs. Limitations: the adapters still cannot create learned identity vectors for taxonomy labels absent from the fixed tokenizer vocabulary unless tokenization itself is deliberately revised.

Method 4: Partial layer fine-tuning

Initialize all eight Waypoint blocks from the checkpoint; freeze early blocks and update selected late blocks plus the classification head. “Freeze blocks 1–6, train blocks 7–8” is an **example candidate configuration**, not a scientifically validated optimal setting.

Method 5: Full transformer fine-tuning

Initialize all blocks from pretrained weights and update every eligible parameter jointly with the classification head. This is most flexible, but for a cohort of 389 runs it has the largest overfitting and model-selection risks.

No new classifier results exist for methods 2–5. LoRA, partial adaptation and full fine-tuning have not yet been executed. It would be inaccurate to write “improved AUC” or “trained transformer” at this stage.

14. Next experiment: LoRA, designed for a fair comparison

The next proposed analysis is supervised LoRA adaptation of the pretrained Waypoint-6m checkpoint. The goal is to test whether adapting the representation to ASD labels improves held-out performance beyond the frozen embeddings and raw-abundance baseline.



The official Waypoint package includes fine-tuning functionality and a LoRA setting. However, its standard split must not be used unchanged if we require the same outer 10-fold protocol. We first inspect the **installed** fine-tuning configuration and confirm the supported classification task, tokenizer behavior, LoRA targets and label mapping.

```
python - <<'PY'
from importlib.resources import files
p = files("waypoint_bio").joinpath("configs/finetune_classification.yaml")
print(p.read_text())
PY
```

One-step next action: Run the config-inspection command on the GPU server and paste the output. Do not begin training with guessed config keys or reuse a single fitted transformer across test folds.

15. Where AGORA2 fits, and what is still pending

AGORA2 is not part of the frozen-embedding classification run. It is a metabolic-model resource, used to explore whether microbial taxa carry annotated metabolic capabilities. Its role is separate from the taxonomic language-model backbone.

AGORA2 preparation	Verified status
Unique original species matched to AGORA2	445 / 827 unique species TaxIDs
Distinct projected genera with AGORA2 match	221 / 340 genus groups
Runs with species-resolved abundance	251 / 389
Runs without species-resolved abundance	138 / 389
Actual metabolite secretion / flux prediction	Not performed

Potential future analysis: map species/genera to AGORA2 metabolic reconstructions, define which functional capacities are supported by those models, and compare carefully annotated functional features with standard abundance features. This does not measure metabolites directly. A genus match does not prove every strain has the same metabolic reactions.

Completed versus not yet completed

Completed	Planned or requiring verification
Corrected 389 × 1,068 matrix and 389 × 355 taxonomic projection	Independent raw source-record duplicate verification
Waypoint vocabulary check: 229 / 355 recognized	Per-sample quality and taxonomic token-order audit
Waypoint-6m embedding extraction: 389 × 256	Frozen MLP / LoRA / partial and full fine-tuning
Four frozen-embedding classifiers, random 10-fold CV	Grouped and project-held-out validation; full metabolic analysis

16. File guide, commands and glossary

Files created or used

File	Meaning
species_model_input_matrix_corrected.tsv	389 × 1,068 corrected input abundance matrix
waypoint_agora_corrected.npz	Corrected matrix, labels, taxonomy mappings, coverage and audit metadata
waypoint_389_taxa_matrix.tsv	389 × 355 taxonomic relative-abundance fraction table
waypoint_389.parquet	Official Waypoint sample/taxonomy input container
waypoint_embeddings_389.csv	389 sample IDs + 256 embedding dimensions
waypoint6m_random10fold_metrics.tsv	Four frozen-embedding classifiers, mean ± fold SD

The two commands already run successfully

```
waypoint prepare-dataset --input waypoint_389_taxa_matrix.tsv \
  --output waypoint_389.parquet --orientation samples_as_rows --no_normalize

waypoint embed --model outpost-bio/Waypoint-6m \
  --data waypoint_389.parquet --output waypoint_embeddings_389.csv \
  --pooling last_token --batch_size 32 --max_length 512 --device cuda
```

Quick glossary

Vocabulary: the set of learned token names and IDs. **Tokenization:** conversion of taxonomic labels into model token IDs. **Embedding:** 256-number vector learned by the pretrained transformer. **Frozen backbone:** pretrained network weights are unchanged. **Fine-tuning:** some pretrained parameters are updated using new labels. **LoRA:** trains compact low-rank adapters. **Outer fold:** held-out test partition. **Mean ± SD:** average metric and variability over folds.

Provenance

Based on the uploaded conversation archive (3.pdf), corrected matrix and NPZ, preprocessing/classification scripts, Waypoint command logs and the four 10-fold results supplied in the conversation. Model architecture/tokenization claims follow the cited workflow described in these materials; exact token order for the worked run was not independently reproduced here from the installed server tokenizer.

Official reference locations: Hugging Face checkpoint `outpost-bio/Waypoint-6m` and Waypoint software repository `github.com/Outpost-Bio/waypoint`.